D 13817 (Pages : 4) Name.....

Reg. No.....

FIRST SEMESTER B.Sc. DEGREE EXAMINATION, NOVEMBER 2016

(CUCBCSS—UG)

Complementary Course

MAT 1C 01—MATHEMATICS

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions.*
Each question carries 1 *mark.*

1. Find $\lim_{t \to 1} \frac{t^2 + t - 2}{t^2 - 1}$.

2. Show that the sum of two continuous functions is continuous.

3. If $\lim_{x \to c} f(x) = 5$ and $\lim_{x \to c} g(x) = -2$, find $\lim_{x \to c} 2f(x)\, g(x)$.

4. Find $K'(-1)$ if $k(z) = \frac{1 - z}{2z}$.

5. If $p = \frac{1}{\sqrt{q+1}}$, find $\frac{dp}{dq}$.

6. What is the derivative product rule ?

7. Does differentiability of a function imply continuity ? Is the converse true ?

8. Find $\lim_{x \to -\infty} \frac{\sqrt{7e}}{x^3}$.

9. Find $\lim_{x \to +\infty} \frac{3x + 7}{x^2 - 2}$.

10. Is the x-axis an aymptote of $f(x) = \frac{1}{x}$?

Turn over

11. Give an example of a non-integrable function on [0, 1].

12. Find the area of the region between $y = 3x^2$ and the x-axis on $[0, b]$.

(12 × 1 = 12 marks)

Section B

Answer any **nine** *questions.*
Each question carries 2 marks.

13. Find $\underset{x \to 9}{\text{Lt}} = \dfrac{\sqrt{x} - 3}{x - 9}$.

14. Prove that $\underset{x \to x_0}{\text{Lt}}\ x = x_0$.

15. Prove that the derivative of a constant function is zero.

16. Find the first derivative of $r = \dfrac{(\theta - 1)\left(\theta^2 + \theta + 1\right)}{\theta^3}$.

17. Find the first four derivatives of $y = x^3 - 7x^2 + 8x - 9$.

18. Find $\underset{x \to \infty}{\text{Lt}} \dfrac{2x^3 + 7}{x^3 - x^2 + x + 7}$.

19. Find a linearization of $f(x) = \sqrt{2 + x}$ at $x = 1$.

20. Find dy if $y = x^3 - 3\sqrt{x}$.

21. If $\int_1^2 f(x)dx = -4$, $\int_1^5 f(x)dx = 6$ and $\int_1^5 g(x)dx = 8$, find $\int_2^5 f(x)dx$ and $\int_1^5 [4f(x) - g(x)]dx$.

22. Define a continuous function.

23. Find the tangent to $y^2 = x$ at (0, 0).

24. State the first derivative test.

(9 × 2 = 18 marks)

Section C

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Applying the definition of limit, show that $\underset{x \to 1}{\text{Lt}} \left(\frac{3x-1}{2} \right)$ is $\frac{1}{2}$.

26. Show that the line $y = mx + b$ is its own tangent at any point $(x_0, mx_0 + b)$.

27. Graph the curve $y = 1/x$. Find its slope at $x = a \neq 0$. When is the slope equal to $-1/4$?

28. Find $\underset{x \to \infty}{\text{Lt}} \; x \sin \left(1/x \right)$.

29. Find the total area between the region $y = -x^2 - 2x, -3 \leq x \leq 2$ and the x-axis.

30. Show that $y = |x|$ is not differentiable at the origin.

31. Show that if n is a positive integer, $\frac{d}{dx}\left(x^n\right) = nx^{n-1}$.

32. State the Mean Value Theorem for definite integrals. Also show that if 'f' is continuous on $[a, b]$, $a \neq b$ and if $\int_a^b f(x)\,dx = 0$, then $f(x) = 0$ atleast once in $[a, b]$.

33. State and prove the fundamental theorem of calculus for the evaluation of definite integrals using anti-derivatives.

$(6 \times 5 = 30$ marks)

Section D

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. (i) Find an equation for the tangent to the curve $y = x + \frac{2}{x}$ at (3, 4).

(ii) Find the second derivative of $p = \frac{q^3 + 3}{12q}\left(\frac{q^4 - 1}{q^3} \right)$.

(iii) At time 't', the position of a body moving along the s-axis is $s = t^3 - 6t^2 + 9t$ metres. Find the body's acceleration each time the velocity is O.

35. (i) Find the critical points of 'f', the intervals in which f is increasing or decreasing and local extrema, if they exist, given $f'(x) = (x-1)^2 (x+2)$.

(ii) Evaluate :

(a) $\underset{x \to \pi/4}{\text{Lt}} \dfrac{\sin x - \cos x}{x - \pi/4}$.

(b) $\underset{x \to \infty}{\text{Lt}} \dfrac{5x^2 - 3x}{7x^2 + 1}$.

36. (i) Find the area between $y = x^4$ and $y = 8x$.

(ii) The circle $x^2 + y^2 = 9$ is rotated about the x-axis to generate a sphere. Find its volume.

(2 × 10 = 20 marks)